DOKUMENTASI DEPLOYMENT WMS ATIN'S COLLECTION

Arsitektur Sistem: Next.js (App Router) ke Cloudflare Workers via OpenNext & Firestore

Tanggal Penyusunan: Oktober 2026 | Environment: Production Live

1. Arsitektur Proyek & Komponen Utama

- Framework: Next.js (App Router, Webpack engine)

- Runtime Host: Cloudflare Workers (OpenNext v1.20+)

- Database: Google Cloud Firestore (Named Database: ai-studio-android-wmsatins-...)

- Skema Auth: Client-Side Verified Auth (menghindari limitasi V8 serverless Cloudflare)

2. Penanganan Masalah Kritis Cloudflare (V8 Engine)

A. Bypassing EvalError (new Function):

Cloudflare Worker melarang dynamic string evaluation. Protobuf.js pada SDK Firestore memicu EvalError saat bootstrapping bundle server. Masalah ini dinetralkan secara otomatis di patch-opennext.js menggunakan closure wrapper (try-catch no-op fallback).

B. Firestore Client Instance:

File lib/firebase.ts wajib menggunakan instance resmi getFirestore(app, databaseId) murni tanpa Proxy wrapper agar kompatibel dengan doc() dan collection().

3. Konfigurasi wrangler.jsonc

```
{
  "$schema": "node_modules/wrangler/config-schema.json",
  "name": "admin-portal",
  "main": ".open-next/worker.js",
  "compatibility_date": "2026-10-13",
  "compatibility_flags": [ "nodejs_compat" ],
  "assets": { "directory": ".open-next/assets", "binding": "ASSETS" }
}
```

4. Konfigurasi patch-opennext.js

Skrip patching dieksekusi otomatis pasca build untuk menambal manifest OpenNext dan runtime protobuf codegen:

```
const targetCodegen = "Function.apply(null,h).apply(null,i)}return Function(b2)()";
const validSafeCodegen = "(() => { try { return Function.apply(null,h).apply(null,i);
code = code.replace(targetCodegen, validSafeCodegen);
```

5. Perintah Build & Deploy Lokal

```
npm run build:worker    # Build Next.js bundle & jalankan patch-opennext.js
npm run deploy:worker   # Upload asset dan deploy worker ke Cloudflare
npx wrangler tail admin-portal  # Live log monitoring
```

6. Deployment Otomatis via GitHub Actions

Simpan kredensial Cloudflare di GitHub Secrets (CLOUDFLARE_API_TOKEN & CLOUDFLARE_ACCOUNT_ID), lalu gunakan konfigurasi .github/workflows/deploy.yml berikut:

```
name: Deploy WMS to Cloudflare Workers
on:
  push:
```

```
    branches: [ main ]
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 20, cache: 'npm' }
      - run: npm ci
      - run: npm run build:worker
      - uses: cloudflare/wrangler-action@v3
        with:
          apiToken: ${{ secrets.CLOUDFLARE_API_TOKEN }}
          accountId: ${{ secrets.CLOUDFLARE_ACCOUNT_ID }}
          command: deploy
```